CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE)

May 2012

CE303 Computer Graphics

Date: 04.05.2012, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain the Basic design of a magnetic deflection CRT. **[04]**

(b) What shadow mask method? Explain the operation of a delta-delta shadow-mask CRT. **[03]**

Q - 2 (a) Draw the architecture of a simple raster graphics system. Explain the Basic video-controller refresh operations. **[07]**

(b) Find the intermediate points of line between the line with endpoints (20, 11) and (30, 22) using Bresenham line drawing algorithm. **[04]**

(c) Write down the steps of Midpoint Circle Algorithm. **[03]**

OR

Q - 2 (a) What is graphics input devices? Explain any two graphics input device thoroughly. **[07]**

(b) Write down steps of DDA line drawing algorithm. **[04]**

(c) Given input ellipse parameters $rx = 8$ and $ry = 6$ calculate points of region 1. **[03]**

Q - 3 (a) Explain the steps required to fill the polygon using flood fill technique. Give the difference between flood fill and boundary fill. **[06]**

(b) What do you mean by composite transformation? How it is useful? **[08]**

OR

Q - 3 (a) Explain shear transformation and reflection transformation with example. **[06]**

(b) Explain the scan line fill algorithm. **[08]**

SECTION – II

Q – 4 Draw a transformation sequence for scaling of an object with respect to a specified fixed position using the scaling matrix S(*sx*, *sy*). Derive the transformation matrix for same. [07]

Q - 5 (a) Explain Cohen-Sutherland line clipping algorithm with example. [07]

(b) Explain Liang-Barsky line clipping algorithm with example. [07]

OR

(b) What is clipping? Define viewport, window and clip window. Explain line clipping and polygon clipping. [07]

Q - 6 Write a short note on ***any TWO***. [14]

1. 3D Scaling
2. Perspective Projection
3. Visible Surface Detection

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